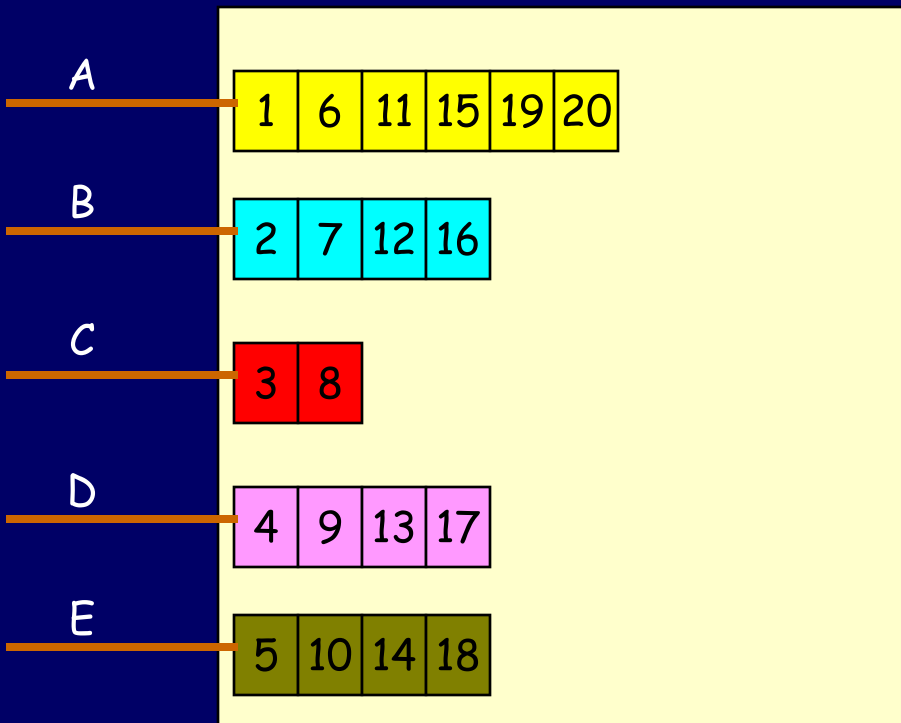
CprE 458/558: Real-Time Systems

Real-Time Networks – WAN
Packet scheduling (Part 2)

Work-conserving vs. Non work-conserving

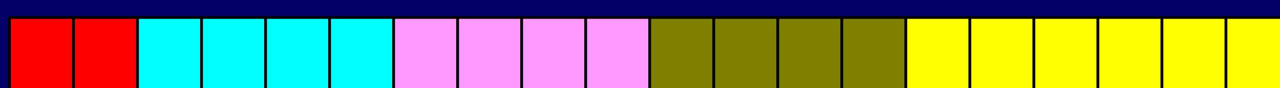
- Work conserving scheduler
  - Never leaves the link idle if there is a packet to be transmitted
  - Offers better link utilization
  - E.g., RR, WRR, WFQ
- Non work-conserving scheduler
  - Associate eligibility time with each packet and transmits packets only when they are eligible
  - Can provide delay-jitter control, easier implementation
  - E.g., HRR

Fair Queuing (FQ) : Byte-by-Byte RR emulation



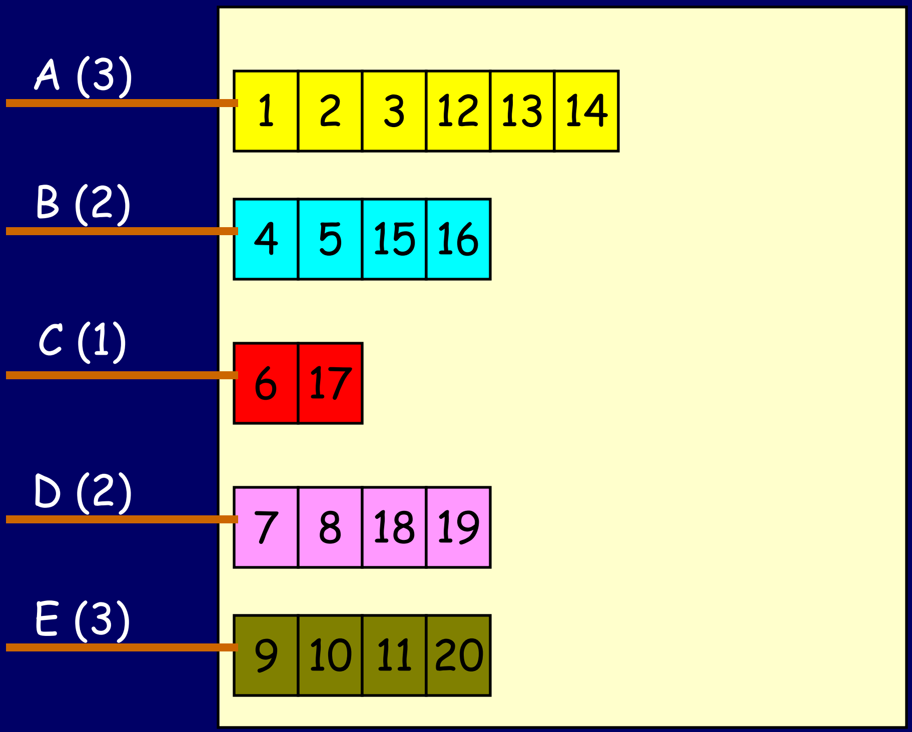
Packet	Finish Time
C	8
B	16
D	17
E	18
A	20

Earliest Finish Time FQ Schedule



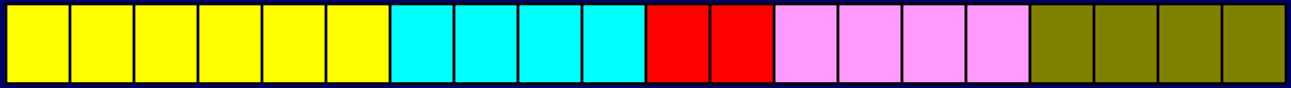
Problem: Gives all the flows the same priority

Weighted Fair Queuing (WFQ)



Packet	Finish Time
A	14
B	16
C	17
D	19
E	20

Earliest Finish Time WFQ Schedule



Finish time/number expressions (1)

- Round Number [R(t)]: number of rounds of service a bit-by-bit round-robin scheduler has completed at a given time.
  - Eg: round number 3.5 means, three full rounds and fourth round is half-way through
- A connection is said to be active if the largest finish number of a packet either in its queue or last served from its queue is larger than the current round number
- Thus, the length of a round, that is, the time taken to serve one bit from each active connection, is proportional to the number of active connections

Finish time/number expressions (2)

- Finish time for an inactive connection is:
  - $F(i, k, t) = R(t) + P(i,k,t) * \varnothing_i$

  - Where F(i, k, t) is the finish number for the kth packet on connection "i'
  - Where, R(t) is the round number
  - P(i,k,t) is the size of the k^{th} packet that arrives on connection "i" at time "t"
  - Where $\varnothing_i$ is the normalized weight ratio of the connection "i".

- Finish time for an active connection is:
  - $F(i, k, t) = F(i, k-1,t) + P(i,k,t) * \varnothing_i$

- The general expression for finish time is:
  - $F(i, k, t) = Max (F(i, k-1,t) , R(t)) + P(i,k,t) * \varnothing_i$

Hierarchical Round Robin (HRR)

- In HRR, there are number of levels, each with a fixed number of slots serviced in a round-robin fashion
- A channel is allocated a given number of service slots at a selected level
- The scheduler cycles through the slots at each level
- The time taken to service all the slots at a given level is called the "<u>frame time</u>" at that level
- The total link bandwidth is partitioned in among these levels
- The key to HRR lies in its ability to give each level a constant share of the link's bandwidth

Hierarchical Round Robin – contd.

- The frame time for level 1, which is the smallest of all the levels, is the basic cycle time.
- If there are n_1 slots in a level 1 frame, then b_1 slots are allocated to higher levels, and the remaining $(n_1 – b_1)$ slots are used for the level 1 connections

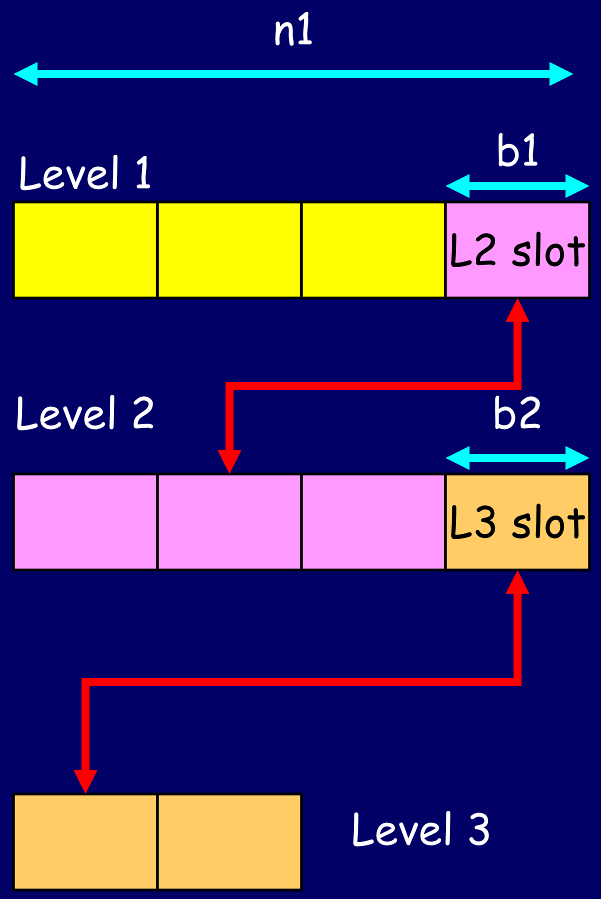
- The frame time for level-1 = $FT_1 = n_1$
- The frame time for level-2 = $FT_2 = (n_1 / b_1) * n_2$

- The frame time for level-I = FT_i =
  $(n_1 / b_1) * (n_2 / b_2) * \ldots (n_{i-1} / b_{i-1}) * n_i$

- Bandwidth allocated to each slot in level i = Link_BW / FT_i
  where Link_BW is the total link bandwidth

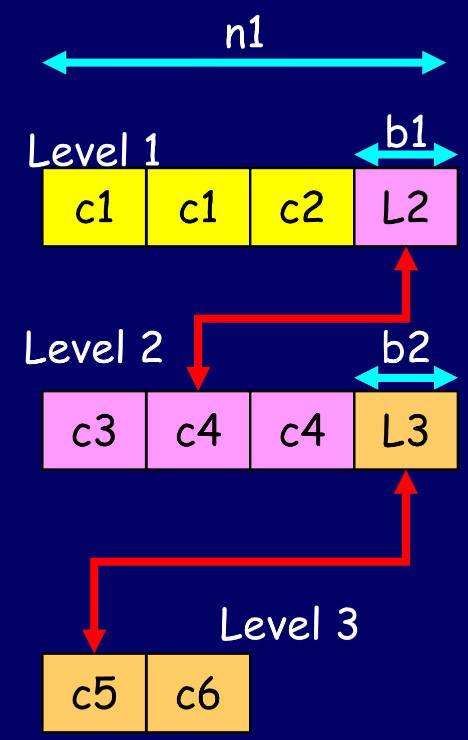
HRR design for a 4Mbps link

Level i	n_i	b_i	FT_i	Slot b/w
1	4	1	4	1 Mbps
2	4	1	16	250 Kbps
3	2	0	32	125 Kbps



HRR – connection allocation example

Channel	Bandwidth need	Level Assigned	# of slots
C1	2 Mbps	1	2
C2	1 Mbps	1	1
C3	250 Kbps	2	1
C4	500 Kbps	2	2
C5	125 Kbps	3	1
C6	100 Kbps	3	1



c1	c1	c2	c3	c1	c1	c2	c4	c1	c1	c2	c4	c1	c1	c2	c5

HRR Schedule up to 16 slots

Real-Time WAN -- Summary

- QoS parameters – bandwidth, delay, delay jitter, packet loss
- Traffic types – CBR and VBR
- Traffic models – Peak rate model, LBAP
- Real-time channel setup
  - QoS routing and Resource reservation
- Data transmission phase
  - Traffic shaping: Leaky bucket, Token bucket
  - Packet scheduling: RR, WRR, WFQ, HRR